Three data paths, one exposure record, reliable health evidence

Worked example with real rows from the synthetic tutorial dataset: person 9 moves from Fauquier County, VA to Lamar County, MS while pregnant

1 CLINICAL PATH

what happened to the person

SOURCE

EHR or claims data at the data partner, standardized to the OMOP CDM: PERSON, OBSERVATION_PERIOD, EPISODE and CONDITION_OCCURRENCE

STANDARDIZED AS

A person_id with clinical time windows: an index date, a pregnancy, a diagnosis. Standard concepts, e.g. Pregnancy 4299535, COPD 255573

EXAMPLE: PERSON 9 (SYNTHETIC)

PERSON
female, born 1981
OBSERVATION_PERIOD
2014−01−01 to 2019−12−31
EPISODE (pregnancy)
2016−05−20 to 2017−02−10 (267 days)

Whole dataset
10,000 persons; 1,086 with a COPD diagnosis

2 LOCATION PATH

where they lived, and when

SOURCE

Addresses stay at the data partner. They are geocoded locally (DeGAUSS, PostGIS, Nominatim or ArcGIS) to a point; only the point and the dates are used downstream

STANDARDIZED AS

OMOP GIS extension: LOCATION (a point) and LOCATION_HISTORY (person, location, start and end date, relationship = Patient Residence 2052496995)

EXAMPLE: LOCATION_HISTORY OF PERSON 9

location 9: 2014−01−01 to 2016−08−14
Fauquier Crossing, VA
Fauquier County, FIPS 51061
location 10009: 2016−08−15 to 2019−12−31
Lamar Crossing, MS
Lamar County, FIPS 28073

The move falls mid−pregnancy and splits one month. 955 of 10,000 persons move once.

3 GEOSPATIAL PATH

what the places were like

SOURCE

Public datasets: CDC monthly PM2.5 (EPA Downscaler), Census TIGER county boundaries, a simulated SES index. Each is described by a catalog entry (JSON−LD metadata)

STANDARDIZED AS

GaiaDB geometry−attribute index: every source is registered the same way (geom_index, attr_index) with its concept, unit, period and source id

EXAMPLE: THE TWO COUNTIES

PM2.5 monthly mean, ug/m3 (source id 23)
Fauquier: Jun 2016 7.65, Jul 2016 8.55
Lamar: Sep 2016 8.82, Oct 2016 10.30
SES index (simulated)
Fauquier 58.6, Lamar 68.2
County polygons (Census TIGER 2023)
Fauquier 51061, Lamar 28073

Whole index: 2,672 counties, 72 months

person_id + clinical window | *location + residence interval* | *polygons + monthly values*

MERGE: THE SPATIAL−TEMPORAL JOIN IN GAIADB (spatial_join_from_catalog)

1 Place: the person's point falls inside a county polygon (ST_Within)
2 Time: the month overlaps the residence interval and is clipped to it
3 Result: one EXTERNAL_EXPOSURE row per person, location and interval, carrying its concept (PM2.5 2052499839), type (Air Quality Database 2052499878), relationship (Within 2052496943) and source id (23)

EXTERNAL_EXPOSURE rows of person 9 around the move (one month, two counties, two rows)

row id	person	location	interval	PM2.5 ug/m3
617	9	9	2016−08−01 to 2016−08−14	6.954 (Fauquier)
618	9	10009	2016−08−15 to 2016−08−31	7.998 (Lamar)

Colors show where each field came from: clinical path, location path, geospatial path. Whole dataset: 720,925 PM2.5 and 10,955 SES rows.

RESEARCH INTEGRATION: THE EXPOSURE RECORD MEETS THE CLINICAL QUESTION

Align to the clinical window

Pregnancy 2016−05−20 to 2017−02−10: 11 exposure rows over 267 days give a day−weighted mean of 8.474 ug/m3. The naive row mean, 8.331, is wrong.

Build cohorts and features

Index 2016−01−01, exposure at index:
high (10 or more ug/m3): 2,717 persons
low (under 8 ug/m3): 3,099 persons
COPD 1,086; negative control CKD 1,384

Estimate the effect

Adjusted for SES, age, sex; county−clustered intervals. COPD slope 0.108 per ug/m3 (simulated truth 0.15); matched hazard ratio 1.26 (1.01−1.57); CKD 0.94 (0.77−1.15)

RELIABLE HEALTH EVIDENCE

Traceable
every row names its source variable (23: CDC county PM2.5)

Verified
720,925 of 720,925 rows match an independent answer key

Quality−checked
4 of 4 deliberately bad rows caught before analysis

Calibrated
negative control stays null; intervals allow for clustering